

happy with the kind of inputs we are providing. Some of the industry leaders have come forward to interact with the academicians and students. Our aim is to build an environment and explore the synergies which are available to meet the kind of skilled personnel that the industry is looking for."

Some other initiatives

While the government of India seems to be taking initiatives for this sector at a slow pace, there are many multinational companies (MNCs) that are pro-actively working to reduce the skill gap. Companies like ROHM, Renesas, NI, Micrel, STMicroelectronics, Texas Instruments provide a free registration to their websites, using which one can get timely news regarding the new tools and kits available, free samples to build projects around these, and design simulators and get online support.

Some of these companies also conduct workshops, webinars, expos and design contests to provide a platform to future engineers so that they can make the best use of the kits and boards and come up with new ideas. They also provide various online training modules based on different kits and design solutions to make things easy. Further, companies are coming up with more and more development boards and providing these at affordable prices to encourage the involvement of the new generation.

There are training institutes, such as EFY Tech Centers, STQC, Young Brain India and CDAC, that provide short-term training on new kits and development boards like Raspberry Pi, Arduino UNO, PIC microcontroller boards and robotics kits in order to further bridge the gap between industry and the academia.

Challenges and suggested solutions

Some of the challenges that the industry experts list out are:

1. **Scattered industry:** They say a

lot of things are being done, but all in some kind of isolation. There is a dire need to connect these dots and bring the industry together.

Solution: Solution to this is to bring out a common platform specifically for the education and training products industry—a platform where people from the industry can interact and move together to impart skill and polish the future engineers of India.

2. **Products without learning outcomes:** A lot of these training and education kits' providers are now producing these kits without paying attention to their learning outcomes. This is not only discouraging potential buyers but also in a way discouraging many engineers from keeping up their self-learning urge.

Solution: The industry gurus suggest that either the government or a few key players from the industry should come forward and lay down some guidelines, specifically for the manufacturing and packaging of these products. It is a sensitive area with regard to its target buyers, and hence, a close vigilance and quality control is required.

Also, inclusion of the technical resources will give a good boost to the growth of the industry.

3. **Awareness amongst students:** One fact that every expert agrees to is the lack of awareness amongst the students for these products. A large number of students are 'ill aware' of the products available in the industry today. What little they know about them is either from their colleagues or teachers.

Solution: Again, a common platform needs to be created where students can find, compare and pick their products according to their specific needs and ideas. Also, workshops, trainings and seminars need to be conducted not only for the students but also for the institutes so that they can adopt such products and more people can join the bandwagon.

4. **High rates of taxes:** Several industry people quoted that high rates

Some Education and Training Kit Providers in India

Suppliers	Type of kits
Adormi Technologies Pvt Ltd	➤ Robotics ➤ Automation ➤ Entertainment ➤ Computer
HPS Electronics	➤ Alarms ➤ Automation ➤ Light Related ➤ Solar ➤ Audio and Musical ➤ Games ➤ Security ➤ Test and Measurement ➤ Programmers ➤ Development boards
Kits'n'Spares	➤ Robotics ➤ GSM Based ➤ PIC Microcontroller Based ➤ PLC Based ➤ VLSI Based ➤ RFID Based ➤ Arduino Based ➤ Freeduino Based ➤ Programmers ➤ Alarms ➤ Development boards
Electro-Kits	➤ Fun and Games ➤ Test and Measurement ➤ Programmers ➤ Security ➤ RF Based boards ➤ Lighting ➤ Development boards
Element14	➤ Development boards
RS Components	➤ Development boards

of taxes is a major bottleneck for the growth of the industry.

Solution: Considering the skill development role of the industry, government can come forward to provide relaxation in tax rates to help the industry grow further, as well as to invite more entrepreneurs.

5. **Lack of industry intervention:** While some initiatives have already been taken in this direction, there is still a gap between what the students need to learn and what the industry is offering.

Solution: As Dr Akhilesh Saurikha, advisor, Ernst & Young, states, "There is scope for Indian players to develop products that suit the curriculum in schools and universities in India and sell customised products.